

Practice Questions

Date : 23-01-2026

Topics:-

1. Function $\rightarrow$ Graph $\rightarrow$ Domain, Range $\rightarrow$ working
2. Composition and combining functions
3. limits of a function
4. Continuous function.

1. Functions :-

- $y = 3x + 5$
- $f(x) = 2x + 3$
- $g(x) = 4 - x$
- $f(x) = -\frac{2}{3}x + 5$

- $f(x) = -3x^2 + 2x - 4$
- $f(x) = 6(x + 1)^2 - 11$
- $f(x) = -5x^2 + 9x - 1$

- $f(x) = \ln(x - 2)$
- $f(x) = e^{x-2}$
- $f(x) = e^x - 3$
- $f(x) = 2\ln(x + 1) - 3$
- $y = \log(3 - x)$

2. Composition

Given: $f(x) = x^2 - 3$ and $h(x) = \sqrt{x+1}$

evaluate each of the following:

- $(f \circ h)(x)$
- $(h \circ f)(x)$
- $(f \circ f)(x)$
- $(h \circ h)(8)$
- $(f \circ h)(4)$

Given $f(x) = 3x - 2$ and $g(x) = \frac{x}{3} + \frac{2}{3}$

evaluate each of the

- $(f \circ g)(x)$
- $(g \circ f)(x)$

Limits :-

$$1. \lim_{x \rightarrow 0} \frac{\sin ax}{\sin bx}$$

$$2. \lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right)$$

$$3. \lim_{x \rightarrow \infty} \left(1 - \frac{1}{x}\right)^x$$

$$4. \lim_{x \rightarrow \infty} \left(\frac{x}{1+x} \right)^x$$

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$$5. \lim_{x \rightarrow \infty} \frac{x^4 - 2x^2 + 6}{x^2 + 7}$$

$$6. \lim_{x \rightarrow \infty} (x - \sqrt{x^2 - a^2})$$

$$7. \lim_{x \rightarrow \infty} \frac{x^2 + 1}{x^{\frac{3}{2}}}$$

$$8. \lim_{x \rightarrow \pm \infty} \frac{5x^3 + 3x^2 - 1}{x - 4x^4}$$

$$9. \lim_{x \rightarrow +2} \frac{\sqrt{4-x^2}}{\sqrt{6-5x+x^2}}$$

$$10. \lim_{x \rightarrow \infty} x + \frac{\sin x}{x}$$

11. let $f(x) = \begin{cases} x^2 - 1 & \text{if } x \leq 2 \\ \sqrt{x+7} & \text{if } x > 2 \end{cases}$
find the limit $f(x)$ as $x \rightarrow 2$

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$$\text{let } f(x) = \begin{cases} \cos x & \text{if } x \leq 0 \\ 1-x & \text{if } x > 0 \end{cases}$$

$$\text{find } \lim_{x \rightarrow 0} f(x)$$

$$\lim_{x \rightarrow 0^-} \frac{x}{x - |x|}$$

$$\lim_{x \rightarrow 1} \frac{x^3 + x^2 - 5x + 3}{x^3 - 3x + 2}$$

last :-

$$g(t) = \begin{cases} t - 2 & t < 0 \\ t^2 & 0 \leq t \leq 2 \\ 2t & t > 2 \end{cases}$$

$$a) \lim_{t \rightarrow 0} g(t)$$

$$b) \lim_{t \rightarrow 1} g(t)$$

$$c) \lim_{t \rightarrow 2} g(t)$$

∴ Continuity :-

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1. Discuss the continuity of $x - |x|$ at $x = 1$.

2. Show that the function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} x & \text{if } x \text{ is irrational} \\ 1-x & \text{if } x \text{ is rational} \end{cases}$$

3. Find c such that the function

$$f(x) = \begin{cases} \frac{1-\sqrt{x}}{x-1} & \text{if } 0 \leq x < 1 \\ c & \text{if } x = 1 \end{cases}$$

is continuous for all $x \in [0, 1]$.

4. Find the value of the constant k if possible, that will make the function continuous everywhere.

$$f(x) = \begin{cases} 7x - 2 & x \leq 1 \\ kx^2 & x > 1. \end{cases}$$

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∴ L' Hospital's Rule :-

$$1. \lim_{x \rightarrow \pi} \frac{\sin^2 x}{\cos 3x + 1}$$

$$2. \lim_{x \rightarrow 0} \frac{e^x - 2\cos x + e^{-x}}{x \sin x}$$

$$3. \lim_{x \rightarrow 0} \frac{\ln(1-x^2)}{\ln(\cos x)}$$

$$4. \lim_{x \rightarrow 0} \frac{\cosh x - \cos x}{x \sin x}$$

$$5. \lim_{x \rightarrow 1} \frac{1 - x + \ln x}{1 - \sqrt{2x - x^2}}$$

$$6. \lim_{x \rightarrow 0} \frac{x \cos x - \ln(1+x)}{x^2}$$

If $\lim_{x \rightarrow 0} \frac{\sin^2 x + a \sin x}{x^3}$ exist, find the values of a and the limit.

$$\lim_{x \rightarrow a} \frac{\ln(x-a)}{\ln(e^x - ea)}$$

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$$\lim_{x \rightarrow 0} \frac{\ln(\tan x)}{\ln x}$$

Asymptotes :-

1. find vertical and horizontal asymptote :-

$$- y = \frac{1}{x}$$

$$- y = \frac{3x^2 + 5}{2x - 1}$$

$$- y = 1 - \frac{2}{x}$$

$$- y = \frac{2x^2 + 3x + 1}{x^2 - 4}$$

and check for
Slant Asymptote
A to b.

$$- y = \frac{x^2 + 1}{x^2 - 2x}$$

$$- y = \frac{2x^2 - 3x + 4}{x + 2}$$

$$- y = \frac{x^3}{x^2 - 1}$$